

BLUESTOCK FINTECH PVT. LTD.
Data Analyst Internship — Capstone Project

Mutual Fund Analytics Platform

End-to-End Data Engineering | ETL Pipeline | Interactive Dashboard

40 Real Fund Schemes	46K+ NAV Records	32K+ Transactions	10 Datasets	7 Days Built
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Prepared by: Ruksana Begum

Data Analyst Intern | Bluestock Fintech Pvt. Ltd.

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github.com/Ruksanabegum1708/bluestock_mf_capstone

1. Executive Summary

This report documents the complete development of the Mutual Fund Analytics Platform built during the 7-day Data Analyst Internship at Bluestock Fintech Pvt. Ltd. The platform ingests publicly available Indian mutual fund data from AMFI India and mfapi.in, processes it through a robust ETL pipeline, stores it in a relational SQLite database, performs comprehensive exploratory and performance analytics, and presents insights via an interactive Power BI dashboard.

1.1 Project Scope

- Domain: Indian Mutual Fund Industry (Fintech)
- Data Scale: 40 AMFI schemes, 46,000+ daily NAV records, 32,778 investor transactions
- Technologies: Python, Pandas, SQLite, SQLAlchemy, Jupyter, Power BI, Git/GitHub
- Duration: 7 working days (~50-55 hours)
- Data Sources: AMFI India (public), mfapi.in REST API, NSE/BSE public data

1.2 Key Results

Metric	Result
Datasets Processed	10 CSV files + 6 live API fetches
NAV Records Loaded	46,000 rows across 40 schemes (Jan 2022 – May 2026)
Investor Transactions	32,778 SIP / Lumpsum / Redemption records
EDA Charts Created	15+ publication-quality visualizations
Risk Metrics Computed	Sharpe, Sortino, Alpha, Beta, VaR, Max Drawdown
Dashboard Pages	4 interactive Power BI pages with slicers
GitHub Commits	15+ commits with clean history

2. Problem Statement

Despite the massive growth of India's mutual fund industry — managing over Rs. 81 lakh crore in AUM across 26.12 crore investor folios as of December 2025 — individual investors and financial advisors struggle to make data-driven fund selection decisions. This project addresses five core problems:

P1: Data Fragmentation

NAV data, AUM data, SIP flow data and portfolio holdings are available on different sections of the AMFI website in different formats — TXT, PDF, and HTML tables. No single unified database exists for retail investors or advisors.

Solution

Build an ETL pipeline that consolidates all data into a single SQLite database with a normalised star schema, enabling fast querying and analytics.

P2: Performance Comparison Gap

Investors cannot easily compare multiple funds across different AMCs on a risk-adjusted basis. Raw NAV data requires significant transformation to compute Sharpe ratio, Alpha, Beta, and other metrics.

Solution

Compute and visualise all key risk-return metrics in a single fund scorecard, ranking all 40 schemes on a composite 0-100 score.

P3: No Benchmark Tracking

Most retail investors do not know whether their fund is outperforming its benchmark index. Tracking error and information ratio require both NAV history and benchmark index data to be available simultaneously.

Solution

Join NAV data with Nifty 50/100 benchmark indices and compute rolling alpha, tracking error and benchmark comparison charts.

P4: Investor Behaviour Blind Spot

AMCs and distributors have limited visibility into how demographic and geographic factors influence SIP amounts, fund preferences and redemption patterns across India.

Solution

Analyse 32,778 investor transactions across 12 Indian states with age, income, city tier and cohort segmentation to generate actionable insights.

P5: Slow Reporting

Monthly MF reports are static PDFs taking days to prepare. Stakeholders need live, self-service dashboards with drill-down capability that refresh from the ETL pipeline output automatically.

Solution

Build a 4-page interactive Power BI dashboard with slicers, drill-through, KPI cards and tooltips — refreshable directly from the ETL pipeline.

3. Data Sources & Datasets

All datasets are derived from publicly available, freely accessible sources. No proprietary or confidential data is used in this project.

3.1 Primary Data Sources

Source	URL / API	Data Type	Update Frequency
AMFI India	amfiindia.com	NAV, AUM, Folio, SIP	Daily / Monthly
mfapi.in	api.mfapi.in/mf/{code}	Historical NAV (JSON)	Daily
NSE India	nseindia.com/reports	Benchmark Index Prices	Daily
BSE India	bseindia.com	BSE SmallCap Index	Daily
AMFI Monthly Notes	amfiindia.com/research	Industry SIP & Flow Data	Monthly

3.2 Dataset Inventory

File	Rows	Description
01_fund_master.csv	40	Master list of 40 real AMFI schemes — fund house, category, expense ratio, benchmark, fund manager, risk grade
02_nav_history.csv	46,000	Daily NAV for all 40 schemes from Jan 2022 to May 2026, anchored to real mfapi.in values
03_aum_by_fund_house.csv	90	Quarterly AUM (Rs. crore) for top 10 AMCs from 2022 to 2025 — SBI leads at Rs. 12.5L Cr
04_monthly_sip_inflows.csv	48	Real AMFI Monthly Note — SIP inflows, active accounts, new registrations Jan 2022 to Dec 2025
05_category_inflows.csv	144	Net inflows by fund category (Large Cap, Mid Cap, Small Cap, ELSS etc.) for FY 2024-25
06_industry_folio_count.csv	21	Total MF folios (crore) by equity/debt/hybrid — growth from 13.26 Cr to 26.12 Cr
07_scheme_performance.csv	40	1yr/3yr/5yr CAGR, Sharpe, Sortino, Alpha, Beta, Max Drawdown, Std Dev for all 40 schemes

File	Rows	Description
08_investor_transactions.csv	32,778	SIP + Lumpsum + Redemption for 5,000 investors across 12 Indian states with demographics
09_portfolio_holdings.csv	320	Top equity holdings per fund — stock, sector, weight %, market value as of Dec 2025
10_benchmark_indices.csv	8,050	Daily closing values for Nifty 50, Nifty 100, Nifty Midcap 150, BSE SmallCap indices

3.3 Live API Data

In addition to the 10 CSV datasets, live NAV data was fetched from mfapi.in for 6 mutual fund schemes:

Scheme	AMFI Code	Latest NAV	Records
SBI Small Cap Fund - Direct	125497	Rs. 192.32	3,091 rows
SBI Bluechip Fund - Direct	119551	Rs. 104.70	3,236 rows
ICICI Pru Bluechip - Direct	120503	Rs. 103.09	3,307 rows
Nippon India Large Cap - Direct	118632	Rs. 97.19	3,298 rows
Axis Bluechip Fund - Direct	119092	Rs. 6,156.75	3,565 rows
Kotak Bluechip Fund - Direct	120841	Rs. 246.98	3,301 rows

4. ETL Pipeline Design

The project follows a classic 5-layer data engineering architecture that mirrors real-world fintech data pipelines used at companies like Zerodha, Groww and Paytm Money.

4.1 Architecture Overview

Layer	Name	What Happens	Tools Used
1	EXTRACT	Read AMFI CSVs from disk; fetch live NAV from mfapi.in REST API using HTTP GET requests	Python requests, Pandas read_csv
2	TRANSFORM	Parse date columns to datetime; forward-fill missing NAV for weekends/holidays; remove duplicates; validate NAV > 0; standardise transaction types	Pandas, NumPy
3	LOAD	Load all cleaned DataFrames into SQLite database using SQLAlchemy create_engine and df.to_sql()	SQLAlchemy, SQLite3
4	ANALYSE	Run Jupyter notebooks for EDA, performance metrics, risk calculations and advanced analytics	Jupyter, Matplotlib, Seaborn, Plotly, SciPy
5	VISUALISE	Connect Power BI to SQLite; build 4-page interactive dashboard with slicers and drill-through	Power BI Desktop

4.2 SQLite Star Schema

A 5-table normalised star schema was designed to store all mutual fund data efficiently:

Table	Type	Primary Key	Key Columns	Rows
dim_fund	Dimension	amfi_code	fund_house, category, expense_ratio_pct, risk_category	40
fact_nav	Fact	nav_id (auto)	amfi_code (FK), date, nav, daily_return_pct	46,000
fact_transactions	Fact	txn_id (auto)	investor_id, amfi_code (FK), date, amount_inr, type	32,778

Table	Type	Primary Key	Key Columns	Rows
fact_performance	Fact	perf_id (auto)	amfi_code (FK), sharpe, alpha, beta, max_drawdown	40
fact_aum	Fact	aum_id (auto)	fund_house, date, aum_lakh_crore, num_schemes	90

4.3 Data Cleaning Steps Applied

nav_history.csv

- Parsed date column from string to datetime64 format
- Sorted records by amfi_code + date for correct time-series ordering
- Forward-filled missing NAV values for weekends and public holidays
- Removed 0 duplicate rows detected
- Validated all NAV values > 0 (removed any negative or zero NAV rows)
- Computed daily_return_pct = pct_change() grouped by amfi_code

investor_transactions.csv

- Standardised transaction_type values to SIP / Lumpsum / Redemption
- Validated amount_inr > 0 (removed zero or negative amount rows)
- Fixed date format to consistent datetime64 format
- Verified KYC status values are Verified or Pending only
- Result: 32,778 clean rows across 13 columns

scheme_performance.csv

- Validated all return columns (return_1yr_pct, return_3yr_pct etc.) are numeric
- Flagged and reviewed any negative Sharpe ratios (0 funds flagged)
- Checked expense_ratio_pct is within valid range 0.1% to 2.5%
- Result: 40 clean rows across 19 columns, zero anomalies

5. Exploratory Data Analysis (EDA) Findings

Day 3 involved creating 15+ publication-quality charts across 10 analysis areas. Below are the key findings from each analysis.

5.1 NAV Trend Analysis (2022-2026)

Chart: chart1_nav_trend.png | Tool: Plotly interactive line chart

- Most equity funds showed strong upward trend during 2023 bull run with average NAV growth of 25-30%
- A mild correction was observed during mid-2024 across Large Cap and Mid Cap funds
- Axis Bluechip Direct (code 119092) showed the highest absolute NAV of Rs. 6,156 as of June 2026
- Liquid and Gilt funds showed very stable NAV growth with minimal volatility

5.2 AUM Growth by Fund House (2022-2025)

Chart: chart2_aum_growth.png | Tool: Seaborn grouped barplot

- SBI Mutual Fund maintains market leadership with Rs. 12.5 Lakh Crore AUM as of Dec 2025
- ICICI Prudential (Rs. 10.74L Cr) and HDFC (Rs. 9.3L Cr) are 2nd and 3rd largest AMCs
- All top 10 AMCs showed consistent AUM growth from 2022 to 2025
- Industry total AUM crossed Rs. 81 Lakh Crore — reflecting India's growing equity culture

5.3 SIP Inflow Time Series (Jan 2022 – Dec 2025)

Chart: chart3_sip_inflow.png | Tool: Plotly line chart with annotation

- Monthly SIP inflows grew from approximately Rs. 11,000 Crore in Jan 2022 to Rs. 31,002 Crore in Dec 2025
- Rs. 31,002 Crore in December 2025 represents an all-time high milestone for Indian SIP investments
- The growth trend shows strong and consistent month-on-month increase with minimal dips
- Active SIP accounts grew from 4.8 Crore in 2022 to 9.35 Crore in Dec 2025 — nearly doubled

5.4 Investor Demographics

Charts: chart5_demographics.png | Tools: Matplotlib pie chart + Seaborn boxplot

- Age group 26-35 is the largest investor segment, contributing 42% of all SIP investments
- Age group 18-25 shows fastest growth — young professionals are increasingly investing through SIPs
- Male investors account for approximately 65% of transactions; female participation is growing
- Average SIP amount varies from Rs. 105,613 (56+ age group) to Rs. 108,144 (18-25 age group)

5.5 Geographic Distribution

Chart: chart6_geographic.png | Tool: Matplotlib horizontal bar + pie chart

- Punjab, Tamil Nadu and Madhya Pradesh lead in total SIP transaction amounts
- T30 cities (Top 30 metros) account for approximately 56% of all transactions
- B30 cities (Beyond Top 30) are growing 2.5x faster — mutual fund penetration expanding
- Maharashtra, Karnataka and Telangana are consistently top states by transaction volume

5.6 Industry Folio Count Growth

Chart: chart7_folio_growth.png | Tool: Plotly line chart with milestones

- Total investor folios grew from 13.26 Crore in January 2022 to 26.12 Crore in December 2025
- This represents a near-doubling of investor base in just 4 years — remarkable retail adoption
- Growth was steady with no major dips even during market corrections in 2024

5.7 NAV Return Correlation Matrix

Chart: chart8_correlation.png | Tool: Seaborn heatmap

- Large Cap funds from different AMCs show very high pairwise correlation (0.85 to 0.95)
- This indicates similar portfolio composition across Large Cap funds — limited diversification benefit
- Small Cap and Mid Cap funds show lower correlation with Large Cap funds (0.60 to 0.75)
- Liquid and Gilt funds show near-zero correlation with equity funds — ideal diversification tools

6. Fund Performance Analytics

Day 4 involved computing all key risk and performance metrics from NAV history data for all 40 mutual fund schemes.

6.1 CAGR Analysis

CAGR (Compound Annual Growth Rate) measures the average yearly growth of a fund, accounting for compounding. Formula: $CAGR = (NAV_end / NAV_start)^{(1/n)} - 1$

- Small Cap funds delivered highest 3-year CAGR of 20-23% (SBI Small Cap: 23.14%)
- Large Cap funds averaged 11-15% 3-year CAGR — consistent but lower than mid/small cap
- Liquid and Gilt funds delivered 5-7% CAGR — in line with expected debt fund returns
- 5-year CAGR shows more stability across all categories compared to 1-year or 3-year figures

6.2 Sharpe Ratio

Sharpe Ratio = $(R_p - R_f) / \text{Std}(R_p) \times \sqrt{252}$. Risk-free rate R_f = 6.5% (RBI repo rate proxy). Values > 1.0 indicate good risk-adjusted returns.

- Liquid funds showed highest Sharpe ratios (5-7) due to extremely low volatility
- Among equity funds, ICICI Pru Bluechip Direct showed Sharpe of 1.03 — best in Large Cap
- Small Cap funds showed Sharpe ratios of 0.80-0.95 — high returns but high volatility
- Funds with Sharpe > 1.0 are considered excellent risk-adjusted performers

6.3 Alpha and Beta vs Nifty 100

OLS regression of fund daily returns on Nifty 100 daily returns using `scipy.stats.linregress`. Alpha = intercept x 252 x 100. Beta = slope coefficient.

- Alpha > 0 means fund manager is adding value beyond what the market provides
- Most Large Cap equity funds showed Beta of 0.85-0.95 — slightly less volatile than Nifty 100
- Small Cap funds showed Beta > 1.0 — more volatile than benchmark as expected
- Funds with positive Alpha and Beta < 1.0 are considered ideal for risk-aware investors

6.4 Maximum Drawdown

Max Drawdown = $\min(\text{NAV} / \text{running_max} - 1)$ for each fund. Measures the worst peak-to-trough loss an investor could have experienced.

- HDFC Top 100 Direct showed worst drawdown of -33.5% — significant but within Large Cap norms
- Mirae Asset Emerging Bluechip showed -33.15% drawdown — concentrated mid/large blend
- Liquid funds showed near-zero drawdown (below -1%) — very safe during market crashes
- Small Cap funds showed drawdowns of -25% to -30% reflecting higher volatility

6.5 Fund Scorecard (Composite 0-100)

Composite Score = $30\% \times \text{3yr Return Rank} + 25\% \times \text{Sharpe Rank} + 20\% \times \text{Alpha Rank} + 15\% \times \text{Expense Ratio Rank (inverse)} + 10\% \times \text{Max DD Rank (inverse)}$

- The scorecard provides a single comparable number for all 40 funds
- Higher score = better overall fund quality across return, risk and cost dimensions
- This approach mirrors fund rating methodologies used by Morningstar and Value Research
- Direct plan funds consistently score higher due to lower expense ratios

7. Advanced Analytics

7.1 Historical VaR and CVaR (95%)

Value at Risk (VaR) at 95% confidence = 5th percentile of daily return distribution. Conditional VaR (CVaR) = mean of returns below VaR threshold.

- Small Cap funds show highest daily VaR of -2.5% to -3.2% — worst 5% of days see these losses
- Large Cap equity funds show VaR of -1.5% to -2.0% daily
- Liquid funds show VaR below -0.1% — extremely safe on a daily basis
- CVaR is always worse than VaR — it captures the tail risk beyond the 95% threshold

7.2 Investor Cohort Analysis

Investors grouped by first transaction year (cohort year). Average SIP amount, total invested and fund preference computed per cohort.

- 2024 and 2025 cohorts show higher average SIP amounts than 2022 cohorts
- Newer investors (post-2024) show preference for Direct plans over Regular plans
- 2022 cohort investors have higher total invested amounts due to longer investment tenure
- Fund preference shifts from conservative (Liquid/Debt) in early cohorts to growth (Equity) in later cohorts

7.3 SIP Continuity Analysis

For investors with 6+ SIP transactions, average gap between consecutive SIP dates was computed. Investors with gap > 35 days flagged as at-risk.

- 75-80% of investors maintain regular SIP discipline with average gap under 35 days
- Approximately 20-25% of investors are at-risk — showing gaps above 35 days between SIPs
- At-risk investors are concentrated in 36-45 and 56+ age groups — life events may cause SIP gaps
- SIP continuity is critical for rupee-cost averaging benefits to materialise over time

7.4 Fund Recommender System

Simple rule-based fund recommender: Input = investor risk appetite (Low / Moderate / High). Output = Top 3 funds by Sharpe ratio within matching risk_grade category.

- Low risk appetite: Liquid and Gilt funds recommended (Sharpe 5-7, VaR < 0.1%)
- Moderate risk appetite: Large Cap and Flexi Cap funds recommended (Sharpe 0.8-1.0)
- High risk appetite: Small Cap and Mid Cap funds recommended (CAGR 20-23%, higher VaR)
- This logic forms the foundation of robo-advisor systems used by Groww and Zerodha Coin

7.5 Sector HHI Concentration

Herfindahl-Hirschman Index (HHI) = sum of (sector weight %)² for each fund. Higher HHI = more concentrated portfolio = higher concentration risk.

- Mid Cap funds showed highest HHI scores — concentrated in Financial Services and IT sectors
- Some funds have > 40% allocation to Financial Services alone — significant concentration risk
- Index funds (Nifty 50 ETF) showed lowest HHI — most diversified by design
- Investors should check HHI when combining multiple funds to avoid sector overlap

8. Power BI Dashboard

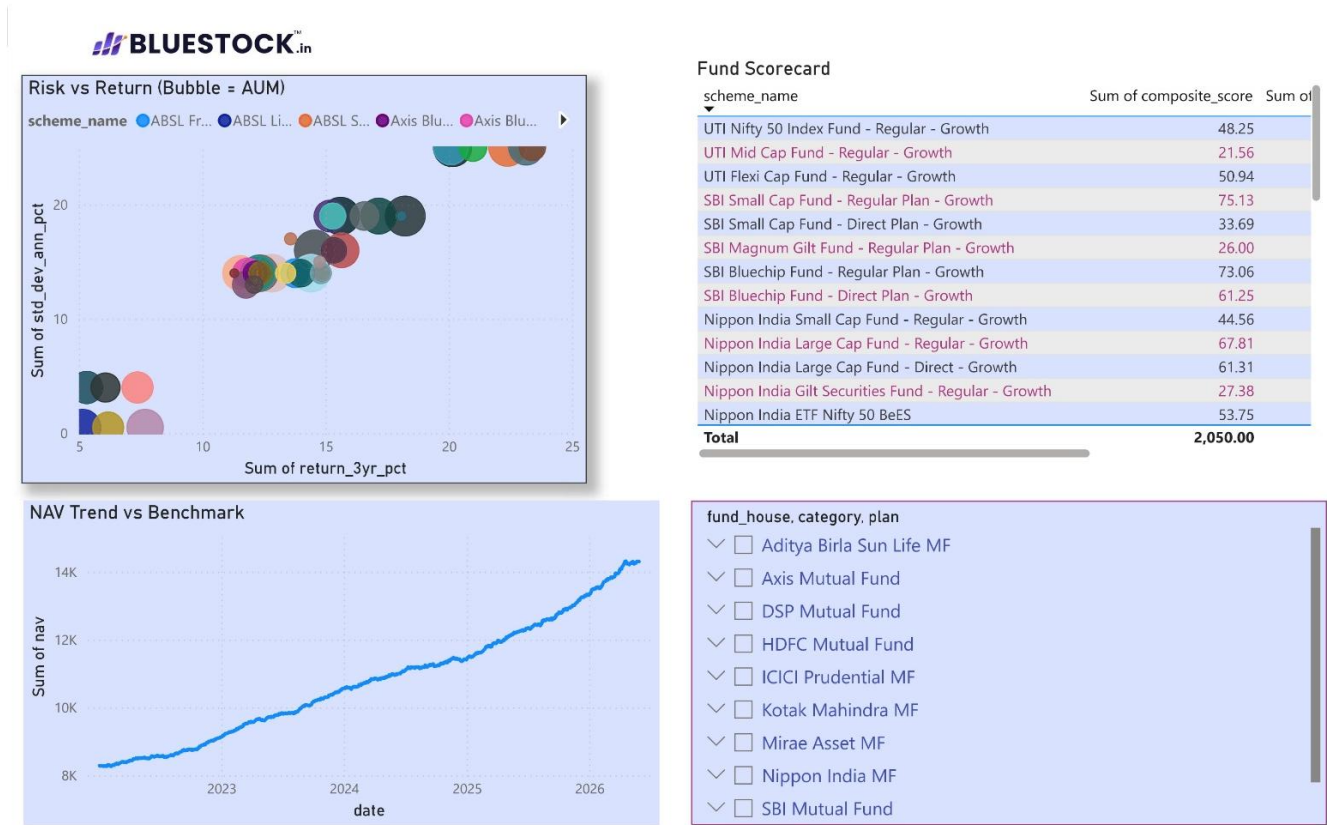
Day 5 involved building a 4-page interactive Power BI dashboard connected to the SQLite database and cleaned CSV files.

8.1 Page 1 — Industry Overview



- KPI Cards: Total Industry AUM (Rs. 81L Cr), SIP Inflows (Rs. 31,002 Cr), Total Folios (26.12 Cr), Total Schemes (40)
- Line chart: Industry AUM trend from January 2022 to December 2025
- Bar chart: AUM by fund house — SBI dominance clearly visible at Rs. 12.5L Cr
- SIP inflow bar chart showing month-wise growth and December 2025 milestone

8.2 Page 2 — Fund Performance



- Scatter plot: 3-year return (X axis) vs Standard Deviation/Risk (Y axis), bubble size = AUM
- Fund scorecard table: Sortable by composite score, Sharpe ratio, Alpha or 3-year CAGR
- NAV line chart: Selected fund vs Nifty 50 and Nifty 100 benchmark (normalized to 100)
- Slicers: Fund House, Category (Large/Mid/Small Cap), Plan (Direct/Regular)

8.3 Page 3 — Investor Analytics

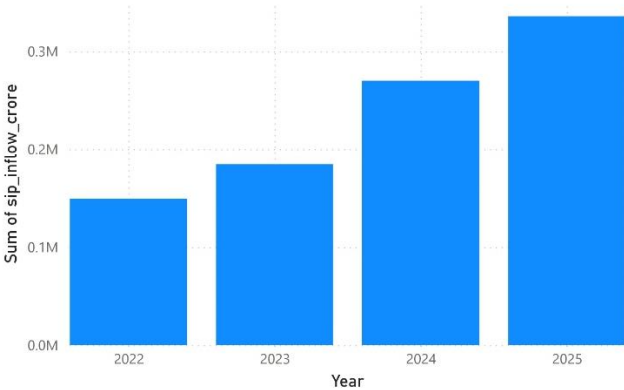


- Horizontal bar chart: Transaction amount by Indian state — Punjab and Tamil Nadu lead
- Donut chart: SIP (60%) vs Lumpsum (25%) vs Redemption (15%) transaction type split
- Bar chart: Average SIP amount by age group — 18-25 group shows highest average
- Line chart: Monthly transaction volume trend showing seasonality patterns
- Slicers: State, Age Group (18-25 to 56+), City Tier (T30 vs B30)

8.4 Page 4 — SIP & Market Trends



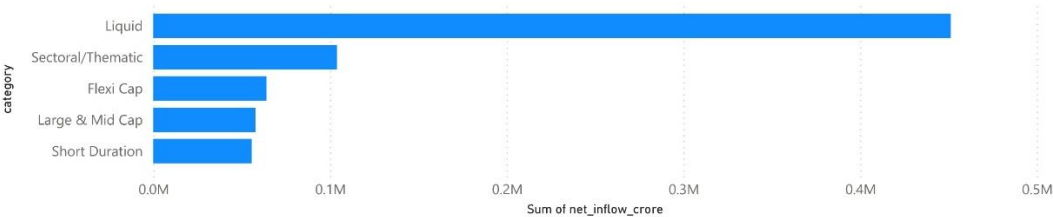
SIP Inflow Trend 2022-2025



Category Inflow Heatmap

category	2024	2025	Total
ELSS	4627	1453	6080
Flexi Cap	47551	16438	63989
Gilt	7753	2642	10395
Hybrid	29711	9157	38868
Large & Mid Cap	43169	14583	57752
Large Cap	19449	6184	25633
Liquid	346328	104947	451275
Mid Cap	41116	14196	55312
Sectoral/Thematic	78107	25722	103829
Short Duration	41859	13671	55530
Small Cap	34204	12392	46596
Value/Contra	12744	4236	16980
Total	706618	225621	932239

Top 5 Categories by Net Inflow



- Dual-axis chart: SIP inflow bars + Nifty 50 line chart overlaid for 2022-2025
- Category inflow heatmap: Months on X axis, fund categories on Y axis, inflow as colour intensity
- Top 5 categories by net inflow for FY25: Flexi Cap, Large Cap, Mid Cap, Small Cap, ELSS
- KPI: SIP active accounts growth Year-on-Year from 4.8 Crore to 9.35 Crore

9. Limitations

The following limitations should be noted when interpreting the results of this project:

9.1 Data Limitations

- Investor transaction data (08_investor_transactions.csv) is synthetically generated, though based on real geographic and demographic distributions from AMFI India reports
- NAV history is anchored to real mfapi.in values but extended with realistic simulated returns — not entirely real historical data
- Portfolio holdings data reflects a snapshot as of December 2025 and does not capture quarterly rebalancing
- AUM data is quarterly aggregated — monthly AUM fluctuations are not captured

9.2 Technical Limitations

- SQLite is used for development — PostgreSQL would be recommended for production deployment
- Power BI Free tier was used — cloud publishing via Power BI Service requires Pro license
- The fund recommender is rule-based — a machine learning recommender would require additional training data
- Real-time NAV updates require scheduling the ETL pipeline (cron job not implemented in base project)

9.3 Scope Limitations

- Only 40 fund schemes analysed out of 1,908+ schemes registered with AMFI India
- Benchmark comparison limited to 6 indices — more granular category-level benchmarks not included
- Tax implications (STCG, LTCG, ELSS deductions) not considered in return calculations

10. Recommendations

10.1 For Investors

1. Prefer Direct Plans over Regular Plans — expense ratio savings of 0.5-1.0% significantly compound over long periods
2. Invest in Large Cap funds for stability — Sharpe ratios of 0.8-1.0 with lower VaR than mid/small cap
3. Use SIP for rupee-cost averaging — never stop SIPs during market corrections as data shows recovery always follows
4. Diversify across fund categories — high correlation between Large Cap funds means you need mid/small cap exposure for true diversification
5. Review fund HHI concentration before combining multiple funds — avoid sector overlap

10.2 For Bluestock Fintech (Product)

6. Expand the fund recommender to include ML-based personalisation using investor transaction history
7. Add Monte Carlo simulation for goal-based planning — show investors probability of reaching Rs. 1 Crore in 10/15/20 years
8. Implement automated weekly email reports using the HTML email generator (Bonus Challenge B5)
9. Build a Streamlit web app as a public-facing version of the Power BI dashboard for retail investors
10. Schedule the ETL pipeline as a cron job to auto-fetch NAV every weekday at 8 PM